AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM254
Managing IT: the why, the what and the how

Final Examination
Spring Semester 2025/2026

Date: 03/June/2026

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 3 Hours

Instructions:

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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PART 1: MCQ [20 Marks]

Each question carries a weight of 2 marks

Question 1: [20 marks] Read the following questions and choose the best answer.

1- Which statement correctly describes a portfolio?
   A. It only contains related projects.
   B. It includes only one program.
   C. It may include related and unrelated projects.
   D. It is a subset of a single project.

2- The work schedule in a project is primarily created using:
   A. The budget reports
   B. Activity bar chart and Resource table
   C. Activity bar chart
   D. Resource table

3- Which of the following is the graphical form of the total costs accumulated over time?
   A. Probability chart
   B. Cumulative project costs
   C. Daily project costs
   D. Stakeholder matrix

4- A project experiences cost overruns due to inaccurate budgeting. This is an example of:
   A. Business risk
   B. Strategic risk
   C. Market risk
   D. Project risk

5- Which development methodology is an example of a combined iterative and incremental approach?
   A. Waterfall
   B. Linear Sequential Model
   C. Big Bang
   D. Agile

6- If a task takes longer than expected due to its complexity, this risk belongs to which category?
   A. Structure
   B. Tasks
   C. Technology
   D. Actors

7- The Closed World Assumption in databases states that:
   A. The database contains incomplete information.
   B. Anything not stored in the database is assumed to be false or unknown.
   C. The database can access external sources for missing information.
   D. All data must be publicly available.

8- What is the result of the intersection between the sets {a, b, c} and {x, y, z}?
   A. {a, b}
   B. {}
   C. {x, y}
   D. {a, x}

9- Which of the following are valid types of backups?
   A. Full
   B. Incremental
   C. Differential
   D. All of the above

10- If a project is falling behind schedule, which of the following actions may help recover the delay?
   A. Replan the activities
   B. Adjust the project scope
   C. Increase resources
   D. All of the above

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PART 2: Short questions [40 Marks]

Question 2: [6 marks]

Explain the concept of the Iron Triangle in project management. Describe its three main priorities and explain how they are related to each other.

Question 3: [6 marks]

Explain the difference between iterative and incremental approaches in software development.

Question 4: [8 marks]

Consider the following relation representing students and exam components:

Exam(**StudentID**, StudentName, CourseID, CourseName)
Exam(1, 'Karim', 'CS101', 'Databases14')
Exam(1, 'Karim', 'CS102', 'Networks14')
Exam(2, 'Walid', 'CS101', 'Databases14')
Exam(3, 'Nour', 'CS103', 'Security14')

a) Explain whether this relation satisfies Second Normal Form (2NF). If it does not, convert it into 2NF. (6 Marks)

b) Justify if the below relation satisfies the first normal form (1NF). If it does not, convert it into 1NF. (2 marks)

Details(StudentID, StudentName, PhoneNumbers)
Details(1, 'Karim', '77123456')
Details(2, 'Walid', '79123456')
Details(3, 'Nour', '89234567')

Question 5: [15 marks]

A library wants to manage information about the books available in its collection. The system should store details about each book, including its title, author, publication year, and price. You are asked to design and manage the database table for this purpose.

a) Create a table named Books with the following structure: (3 marks)
- BookID (integer, primary key)
- Title (varchar(60), not null)
- Author (varchar(50))
- PublishedYear (integer, must be greater than or equal to 1900)
- Price (double)

b) Insert the following data into the Books table: (3 marks)

(1, 'Database Systems', 'Carla', 2016, 55.00)
(2, 'Advanced Databases', 'Carla', 2018, 60.00)
(3, 'Operating Systems', 'Karim', 2014, 70.00)
(4, 'Artificial Intelligence', 'Farid', 2020, 80.00)

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c) Write an SQL statement to display all books published after 2015. (2 marks)

d) Update the price of the book "Operating Systems" to 75.00. (3 marks)

e) Write an SQL query to display the number of books per author. (2 marks)

f) Create a new SQL user named user1 with password 'Default'. (2 marks)

Question 6: [5 marks]

Consider the following two tables:

Table 1: Students

StudentID	Name	Country
1	Karim14	Lebanon
2	Walid14	Oman
3	Nour14	Lebanon

Table 2: ExamResults

StudentID	ExamType	Grade
1	MTA	20
2	Final	40
3	TMA	15

Write the output of each of the following SQL queries:

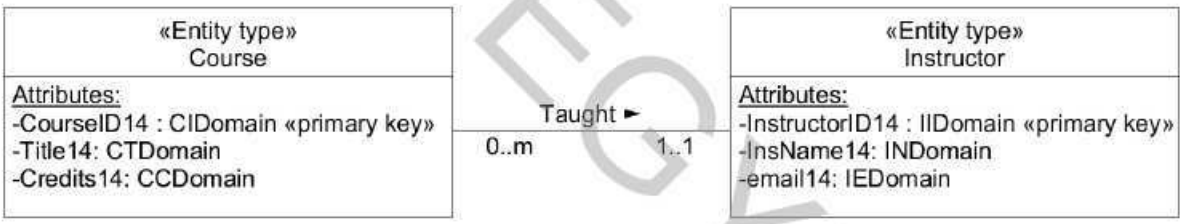
a. SELECT Name FROM Students WHERE Country = 'Lebanon';

b. SELECT Name, Grade FROM Students S JOIN ExamResults E ON S.StudentID = E.StudentID WHERE Grade >= 20;

Part 3: Problem Solving Questions [40 Marks]

Question 7: [25 marks]

The below E-R diagram shows the relationship **Taught** between the Course and Instructor entity types.



a) Write the textual notation for each entity type, including primary keys and attributes. (3 Marks)

b) What is the optionality and multiplicity of **Course** and **Instructor** in the **Taught** relationship? (4 Marks)

c) Draw the relation-for-relationship representation from the original E-R diagram. (6 Marks)

d) Draw the posted-key representation from the original E-R diagrams. (5 Marks)

e) Clarify when the posted-key representation is not suitable. (2 marks)

f) Create an Entity-Relationship Diagram (ERD) that represents the relationships described Below: (5 Marks)

Conceptual Model

- Entities: Student, Course, Instructor.

Relationships:

- Each student may register for one course.
- Each course may have one or more students registered.
- Each course must be taught by exactly one instructor.
- Each instructor may teach one or more courses.

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Question 9: [15 marks]

The Development Team is planning a new website launch. The project involves several key activities with specified durations and dependencies.

Activity	Task Description	Duration (days)	Predecessor(s)
A	Gather user requirements	4	—
B	Set up development environment	3	—
C	Create system design	5	A, B
D	Develop database module	4	B
E	Prepare content and UI assets	2	B
F	Integrate design with database	3	C, D
G	Finalize UI content	3	E
H	Final testing and deployment	2	F, G

a) Draw the Activity-on-Node (AoN) diagram (3 marks)
b) Calculate and fill the duration, earliest start, and earliest finish values for each node. (6 marks)
c) Calculate and fill the latest start, float, and latest finish values for each node. (6 marks)

End of questions

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